```
% Drone Thrust-to-Weight Analysis

MaterialNames = ["Carbon Fiber Composite (CFRP)"; "Aluminum Alloy"; "Fiberglass
Composite"; "PLA Plastic"; "ABS Plastic"; "Wood"];
for i = 1:6

    % Constants
    ThrustPerMotor = 1; % In kg
    Gravity = 9.81; % In m/s^2
    BaseDroneMass = 1; % In kg
    NumberOfMotors = 4;
    Volume = 0.000014344412;
    Density = materials(i).rho_kg_m3;


    % Arm Mass
    ArmMass = Volume*Density; % In kg.


    % Payload
    Payload = 0.5; % Minimum Required Payload


    % Total Drone Mass
    TotalMass = BaseDroneMass + (NumberOfMotors*ArmMass) + Payload;


    % Total Thrust
    TotalThrust = (NumberOfMotors*ThrustPerMotor);


    % Thrust-To-Weight
    ThrustToWeightRatio = TotalThrust/TotalMass;


    % Maximum Allowable Mass
    MaxTotalMass = TotalThrust/2; % Incorporates the 2:1 ratio


    % Maximum Payload
    CurrentMaxPayload = MaxTotalMass - BaseDroneMass - (NumberOfMotors*ArmMass); %
Max payload and still satisfies the 2:1 thrust-to-weight requirement

    % Store Results
    MaxPayloads(i) = CurrentMaxPayload;
    ThrustRatios(i) = ThrustToWeightRatio;

    % Show Results
    fprintf("\nMaterial: %s\n", MaterialNames(i));
    fprintf("Total Drone Mass: %.3f kg\n", TotalMass);
```

```
    fprintf("Maximum Payload: %.3f kg\n", CurrentMaxPayload);
    fprintf("Thrust-to-Weight Ratio: %.2f\n", ThrustToWeightRatio);


    % Check Requirements
    if ThrustToWeightRatio >= 2 && CurrentMaxPayload >= 0.5
        disp("Design meets the requirements")
    else
        disp("Design does not meet the requirements")
    end

end
```

```
Material: Carbon Fiber Composite (CFRP)
Total Drone Mass: 1.592 kg
Maximum Payload: 0.908 kg
Thrust-to-Weight Ratio: 2.51
Design meets the requirements
Material: Aluminum Alloy
Total Drone Mass: 1.655 kg
Maximum Payload: 0.845 kg
Thrust-to-Weight Ratio: 2.42
Design meets the requirements
Material: Fiberglass Composite
Total Drone Mass: 1.609 kg
Maximum Payload: 0.891 kg
Thrust-to-Weight Ratio: 2.49
Design meets the requirements
Material: PLA Plastic
Total Drone Mass: 1.572 kg
Maximum Payload: 0.928 kg
Thrust-to-Weight Ratio: 2.54
Design meets the requirements
Material: ABS Plastic
Total Drone Mass: 1.560 kg
Maximum Payload: 0.940 kg
Thrust-to-Weight Ratio: 2.56
Design meets the requirements
Material: Wood
Total Drone Mass: 1.534 kg
Maximum Payload: 0.966 kg
Thrust-to-Weight Ratio: 2.61
Design meets the requirements
```

```
ResultsTable = table(MaterialNames, MaxPayloads', ThrustRatios','VariableNames',
{'Material', 'MaxPayload_kg','ThrustToWeightRatio'});

disp(ResultsTable)
```

Material	MaxPayload_kg	ThrustToWeightRatio
"Carbon Fiber Composite (CFRP)"	0.9082	2.5129
"Aluminum Alloy"	0.84508	2.417
"Fiberglass Composite"	0.89098	2.486
"PLA Plastic"	0.92828	2.545
"ABS Plastic"	0.93975	2.5637
"Wood"	0.96557	2.6068